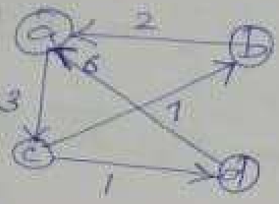
Floyd's Algorithm - All pairs shortest path problem



Weight matrix is created by marking distance of direct edges and if there are no self loops, diagonal entries are 0. Not only that, if there are no direct edge connecting 2 Vertices, its distance value is marked as ∞.

$$W = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & \infty & 3 & \infty \\ 2 & 2 & 0 & \infty & \infty \\ 3 & \infty & 7 & 0 & 1 \\ 4 & 6 & \infty & \infty & 0 \end{array} = D^{(0)}$$

We can copy 1^{st} row and col as such in $D^{(1)}$. We can also mark all diagonal entries as 0

$$D^{(1)} = \begin{bmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & 5 & \infty \\ \infty & 7 & 0 & 1 \\ 6 & \infty & 9 & 0 \end{bmatrix}$$

Consider 'a' as the intermediate Vertex

$$d_{ij}^{k} = \min \left\{ d_{ij}^{k-1},\ d_{ik}^{k-1} + d_{kj}^{k-1} \right\}$$

for $k \geq 1$, $d_{ij}^{(0)} = w_{ij}$

$$d_{23}^{1} = \min\left\{ d_{23}^{0},\ d_{21}^{0} + d_{13}^{0} \right\}$$

$$d_{23}^{1} = \left\{ \infty,\ 2+3 \right\} = 5$$

$$d_{24}^{(1)} = \min\{d_{24}^{0}, d_{21}^{0} + d_{14}^{0}\}$$
$$= \min\{\infty, 2+\infty\} = \infty$$

$$d_{32}^{(1)} = \min\{d_{32}^{0}, d_{31}^{0} + d_{12}^{0}\}$$
$$= \min\{7, \infty+\infty\} = 7$$

$$d_{34}^{(1)} = \min\{d_{34}^{0}, d_{31}^{0} + d_{14}^{0}\}$$
$$= \min\{1, \infty+\infty\} = 1$$

$$d_{42}^{(1)} = \min\{d_{42}^{0}, d_{41}^{0} + d_{12}^{0}\}$$
$$= \min\{\infty, 6+\infty\} = \infty$$

$$d_{43}^{(1)} = \min\{d_{43}^{0}, d_{41}^{0} + d_{13}^{0}\}$$
$$= \min\{\infty, 6+3\} = 9$$

$$D^{(2)} = \begin{array}{c c} & \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & 5 & \infty \\ 9 & 7 & 0 & 1 \\ 6 & \infty & 9 & 0 \end{bmatrix} \end{array}$$

Consider 'b' as the intermediate vertex

$$D^{(3)} = \begin{bmatrix} 0 & 10 & 3 & 4 \\ 2 & 0 & 5 & 6 \\ 9 & 7 & 0 & 1 \\ 6 & 16 & 9 & 0 \end{bmatrix}$$

Consider 'c' as the intermediate vertex

$$D^{(4)} = \begin{array}{c c} & \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 0 & 10 & 3 & 4 \\ 2 & 0 & 5 & 6 \\ 7 & 7 & 0 & 1 \\ 6 & 16 & 9 & 0 \end{bmatrix} \end{array}$$

Consider 'd' as the intermediate vertex.

Shortest Distances between all pairs are listed below.

a–b : 10	b–a : 2	c–a : 7	d–a : 6
a–c : 3	b–c : 5	c–b : 7	d–b : 16
a–d : 4	b–c : 6	c–d : 1	d–c : 9

(Ans)